The Universal Semantic Manifold:
Explicit Semantic Geometry from Validation to Scale

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Abstract

Current generative AI is *token-first*: it predicts the next symbol from a flat vocabulary, and meaning emerges only as a statistical side-effect of token co-occurrence. We argue this order is backwards. Meaning should come first; tokens should come last.

We introduce the *Universal Semantic Manifold* (USM), a Riemannian framework that makes semantic structure an explicit geometric object. Every concept occupies a point on the Poincaré ball $\mathbb{B}^d$ where geodesic distance encodes relatedness, radial depth encodes abstraction level, typed relation maps encode causal and taxonomic structure, and a non-parametric density field distinguishes well-attested knowledge from uncertain territory.

We present two empirical studies. At *validation scale* (100K ConceptNet triples, 85K concepts, d=512), the Poincaré model outperforms an identically-trained Euclidean baseline on knowledge graph link prediction (+25% MRR, +27% Hits@10) and hierarchy depth accuracy (+16 pp), confirming that curvature captures semantic structure that flat geometry cannot. At *cloud scale* (400K triples, 210K concepts, d=1024, NVIDIA A100), the hyperbolic advantage disappears: the Euclidean baseline wins on every metric, including a catastrophic collapse of hierarchy depth ordering (20% vs. 99%). Analysis reveals that the failure is optimisational, not geometric: the Poincaré projection collapsed to a constant-radius shell, and geodesic-based contrastive learning did not converge. We identify learnable curvature, Riemannian-aware gradient control, and curriculum training as necessary ingredients for scaling hyperbolic embeddings, and leave their implementation as open work.

1 Introduction

1.1 The Token-First Paradigm and Its Limits

The dominant paradigm in generative AI is *token-first* generation. A large language model receives a sequence of tokens and predicts the next one. Knowledge, insofar as it exists, is an emergent property of billions of token co-occurrence statistics compressed into transformer weights [Vaswani et al., 2017]. This produces extraordinary fluency, but carries a structural problem: *the model has no explicit representation of ideas*.

Consider “a dolphin is a mammal.” A token-first model represents this as a statistical regularity: the sequence `dolphin--is--a--mammal` has high probability under the training distribution. The concept that dolphins belong to the mammalian class, with all its implications for breathing, lactation, warm-bloodedness, and evolutionary descent, exists nowhere as a structured object. It is distributed across billions of parameters, entangled with syntactic patterns, language-specific morphology, and tokenisation artifacts.

This entanglement has three consequences:

1. **Hallucination is structurally undetectable.** A model cannot distinguish “dolphin is a mammal” (true) from “dolphin is a reptile” (false) at the representation level. Both are token sequences with associated probabilities.

2. **Reasoning is simulated, not computed.** Chain-of-thought prompting produces reasoning-like tokens, but the underlying computation remains next-token prediction. There is no mechanism for structured inference over typed relations (IS-A, CAUSES, PART-OF).
3. **Cross-modal and cross-lingual transfer is indirect.** An image of a dolphin, the English word "dolphin," and the French word "dauphin" should all point to the same idea. In token-first systems, alignment is achieved through massive paired data and contrastive training [Radford et al., 2021], but the idea itself has no unified representation.

1.2 The Idea-First Alternative

We propose inverting the pipeline (Figure 1). Instead of generating tokens and hoping meaning emerges, we construct a geometric space where ideas exist as explicit objects, and then decode from ideas to tokens only at the final output step.

Definition 1.1 (Idea-First Generation)**.** A generation system is *idea-first* if it:

1. Maintains a structured representation space where each point corresponds to a semantic concept with measurable geometric properties (depth, neighbourhood, density);
2. Performs reasoning as geometric operations (composition, relation-following, navigation) in this space;
3. Decodes to a surface form (text, image, speech) only as a final rendering step.

The critical distinction from token-first systems is that the statistical structure being learned is not the co-occurrence of tokens but the *logic of ideas*: which concepts entail which others, what causes what, what is part of what, and how abstract or specific each concept is. These are not implicit patterns to be extracted post-hoc from token statistics; they are the training objective itself.

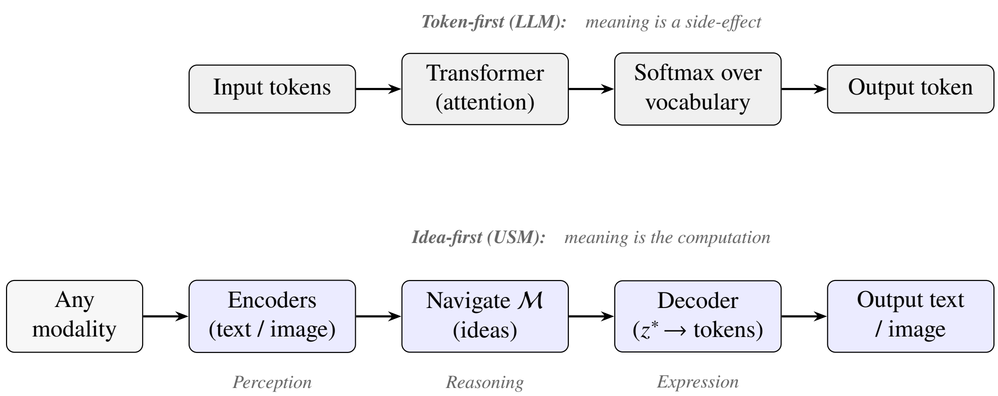

Figure 1: Token-first generation (top) vs. idea-first generation (bottom). In the USM, reasoning operates in the semantic manifold $\mathcal{M}$ before any token is produced.

1.3 Contributions

1. **A geometric framework for explicit semantic structure.** The USM represents concepts on the Poincaré ball where hierarchy, relations, composition, and density are geometrically encoded rather than implicitly learned.
2. **Validation-scale evidence for the geometry of meaning.** At 85K concepts and d=512, the Poincaré model outperforms an identical Euclidean baseline on link prediction (MRR +25%, Hits@10 +27%) and hierarchy depth (+16 pp), providing direct evidence that curvature captures semantic structure that flat geometry cannot.
3. **An honest scale-up and its failure.** At 210K concepts and d=1024, the hyperbolic advantage disappears entirely. We report these negative results transparently and diagnose the root causes:

radius collapse, gradient distortion near the boundary, and insufficient hierarchy-aware training curriculum.

4. **A roadmap for scaling hyperbolic semantics.** We identify learnable curvature, Riemannian gradient clipping, and curriculum-based training as necessary (and currently missing) ingredients, framing the scaling problem as open research.

2 The Logic of Ideas vs. the Statistics of Tokens

The central thesis of this paper is that ideas have a *statistical logic* that is distinct from, and more fundamental than, the statistical regularities of tokens. We make this distinction precise.

2.1 Token Statistics: What Transformers Learn

A transformer trained on next-token prediction learns a conditional distribution $p(x_{t+1} \mid x_1, \ldots, x_t)$ over a discrete vocabulary. The knowledge encoded in the model is the statistical structure of this distribution: which tokens tend to follow which other tokens, conditioned on context. This produces fluent text, and some degree of factual accuracy emerges from the scale of training data [Brown et al., 2020].

But token statistics are fundamentally surface-level. The statement "a dog is an animal" and "un chien est un animal" express the same idea, but their token statistics are entirely different. The statement "fire causes smoke" and "smoke is caused by fire" have overlapping tokens but encode a directed causal relation that is not captured by co-occurrence alone.

2.2 Idea Statistics: What Should Be Learned Instead

Ideas have their own statistical structure, one that is language-independent, modality-independent, and explicitly relational:

- **Hierarchical statistics.** The concept *golden retriever* is more specific than *dog*, which is more specific than *animal*. This depth relationship is a property of the concepts themselves, not of any token sequence. In a taxonomy, the branching factor and depth of specialisation follow regular statistical patterns.
- **Relational statistics.** The relation CAUSES appears between certain pairs of concepts (fire $\rightarrow$ smoke, rain $\rightarrow$ flooding) but not others (fire $\rightarrow$ justice). The distribution of which concept-pairs participate in which relations is a statistical structure over ideas, not tokens.
- **Compositional statistics.** Compound concepts such as "hot dog" and "fire engine" have meanings that are non-linear functions of their parts. IS-A compositions are typically monotonic (a red dog IS-A dog IS-A animal), while CAUSES compositions can be transitive (stress causes anxiety causes insomnia).
- **Entailment statistics.** "All dogs are animals" entails "some animals exist." "It is raining" contradicts "the sky is clear." These logical relationships between ideas have distributional structure that is entirely independent of the surface tokens used to express them.

2.3 Why Geometry Is the Right Representation

We argue that the natural representation for idea statistics is not a discrete probability table but a continuous geometry:

1. **Hierarchy needs curvature.** A taxonomy with k children per node has k^n concepts at depth n. Euclidean space grows polynomially with radius; hyperbolic space grows exponentially. This is not a metaphor; it is a mathematical fact [Nickel and Kiela, 2017]. Flat geometry distorts hierarchies; curved geometry preserves them.

2. **Relations need typed maps.** IS-A (asymmetric, transitive) has fundamentally different algebraic structure from SIMILAR-TO (symmetric, non-transitive). Geometry provides the language to express this: IS-A maps points outward on the ball (increasing depth); SIMILAR-TO preserves depth while rotating within a level.
3. **Composition needs non-linearity.** The meaning of "fire engine" is not the average of "fire" and "engine." A bilinear operator in curved space can capture emergent compound meaning that no additive model can.
4. **Density needs a manifold.** Some ideas are well-attested ("water is wet") and others are speculative ("consciousness is quantum"). The density of training signal at different manifold positions provides a natural measure of epistemic confidence.

The USM's proposition is to build a space where the geometry *is* the logic of ideas, then navigate that space at inference time. Tokens are a decoding format.

3 Background and Related Work

3.1 Hyperbolic Embeddings

Nickel and Kiela [2017] demonstrated that hierarchical data can be embedded with far less distortion in the Poincaré ball $\mathbb{B}^d = \{z \in \mathbb{R}^d : \|z\| < 1\}$ than in Euclidean space. The key insight is that hyperbolic volume grows as $e^{(d-1)r}$ with radius r, matching the exponential growth of trees. Ganea et al. [2018] extended this to neural networks. Desai et al. [2023] applied hyperbolic geometry to image-text representations (MERU), showing that entailment cones in the Poincaré ball model visual-semantic hierarchies. The USM builds on all of these but learns relational structure, compositional operators, and density jointly with the geometry.

3.2 Knowledge Graph Embeddings

TransE [Bordes et al., 2013] models relations as translations: $z_h + z_r \approx z_t$. MurE [Balažević et al., 2019] extends this to hyperbolic space. The USM replaces fixed translation vectors with learned non-linear maps ϕ_r, enabling representation of relations like CAUSES that are poorly modelled by vector addition.

3.3 Contrastive and Cross-Modal Learning

CLIP [Radford et al., 2021] aligns vision and language via InfoNCE [van den Oord et al., 2018]. The USM adopts the same contrastive framework but operates in geodesic distance rather than cosine similarity, and extends it to four-language cross-lingual alignment and a joint text-vision manifold.

3.4 Order Embeddings and Entailment

Vendrov et al. [2016] proposed order embeddings for visual-semantic hierarchies. HypStructure [HypStructure, 2024] at NeurIPS 2024 developed structured hyperbolic representations. We adopt the Vendrov formulation extended to the Poincaré tangent space and augmented with a margin-based depth ranking loss.

4 The Universal Semantic Manifold

4.1 The Manifold

Definition 4.1 (Universal Semantic Manifold)**.** The Universal Semantic Manifold is a smooth Riemannian manifold $(\mathcal{M}, g)$ equipped with:

- A metric tensor $g_{ij}(z)$ inducing geodesic distance $d_{\mathcal{M}}(z_1, z_2)$

- A non-parametric density field $\rho(z)$ encoding epistemic confidence
- A family of typed relation maps $\{\phi_r : \mathcal{M} \to \mathcal{M}\}_{r \in \mathcal{R}}$
- A typed bilinear composition operator $\phi(z_A, r, z_B) \to z_C$

We instantiate $\mathcal{M}$ as the Poincaré ball $\mathbb{B}_c^d = \{z \in \mathbb{R}^d : c\,\|z\|^2 < 1\}$ with curvature $c = 1$. For numerical stability, we use the direct arcosh formula for geodesic distance:

$$d_{\mathcal{M}}(z_1, z_2) = \operatorname{arcosh}\left(1 + \frac{2\,\|z_1 - z_2\|^2}{(1 - \|z_1\|^2)(1 - \|z_2\|^2)}\right)$$

which is perfectly symmetric and avoids the numerical asymmetry in implementations based on Möbius addition. All points are clamped to $\|z\| \leq 1 - \varepsilon$ with $\varepsilon = 10^{-5}$ after every exponential map operation.

4.2 Semantic Depth as Radial Position

The geodesic distance from the origin $d_{\mathcal{M}}(z, \mathbf{0})$ measures concept specificity (Figure 2):

- Near the origin: general, abstract concepts (animal, object, event).
- Near the boundary: specific, concrete concepts (golden retriever, Toyota Corolla, the 2024 Paris Olympics).

This is enforced via the hierarchy loss (Section 4.5): fine-grained concepts are pushed deeper than their coarse superclasses by a learned margin.

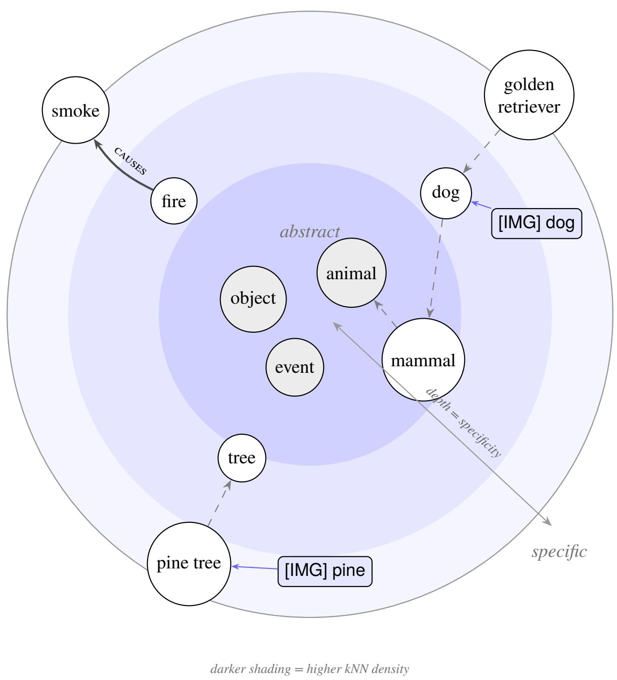

Figure 2: A 2-D cross-section of the Poincaré ball. Abstract concepts (animal, event) cluster near the origin; specific concepts (golden retriever, pine tree) lie near the boundary. Dashed arrows indicate IS-A relations (increasing depth); solid arrows indicate CAUSES. Blue rectangles are image embeddings that co-locate with their text counterparts, illustrating cross-modal alignment. Shading indicates kNN density: well-attested regions are darker.

4.3 The Encoder Family

Each input modality has a dedicated encoder projecting to the manifold.

Text encoder. A frozen `all-MiniLM-L6-v2` [Reimers and Gurevych, 2019] backbone produces 384-dimensional representations, followed by a trainable projection $\mathbb{R}^{384} \to \mathbb{R}^d$ with LayerNorm and the exponential map to $\mathbb{B}^d$. The backbone is frozen to isolate geometric learning from feature learning.

Vision encoder. A frozen CLIP vision backbone (ViT-B/32 or ViT-L/14 depending on available compute) extracts image features, followed by a trainable projection $\mathbb{R}^{d_{\text{CLIP}}} \to \mathbb{R}^d$ and exponential map. The CLIP backbone is frozen; only the projection learns to map visual features onto the semantic manifold.

Both encoders share the same target manifold. After training, an image of a dolphin and the text "dolphin" should occupy nearby points on $\mathcal{M}$.

4.4 The Compositional Operator

Definition 4.2 (Typed Bilinear Compositional Operator)**.** Given concepts $z_A, z_B \in \mathcal{M}$ and relation type r, the compositional operator produces a compound concept:

$$\phi(z_A, r, z_B) = \sigma\Big(W_r\, a + U_r^{\text{eff}}\, b + V_r\, (a \odot b) + b_r\Big)$$

where $a = \log_0(z_A)$, $b = \log_0(z_B)$, $\{W_r, U_r, V_r, b_r\}$ are per-relation learnable parameters, and the output is projected back to $\mathcal{M}$ via $\exp_0$.

The $V_r\,(a \odot b)$ term captures emergent compound meaning: properties that neither argument alone possesses. Symmetry is enforced structurally via $U_r^{\text{eff}} = s_r \cdot W_r + (1 - s_r) \cdot U_r^{\text{raw}}$, where $s_r = 1.0$ for SIMILAR-TO (perfectly symmetric), $s_r = 0.0$ for directed relations (IS-A, CAUSES, PART-OF), and $s_r = 0.1$ for ANTONYM (near-symmetric with learned flip).

4.5 The Training Objective

The full objective is:

$$\mathcal{L} = \mathcal{L}_{REL} + \mathcal{L}_{CL} + \lambda_{ent}\, \mathcal{L}_{ENT} + \lambda_{comp}\, \mathcal{L}_{COMP} + \mathcal{L}_{CM} + \lambda_{hier}\, \mathcal{L}_{HIER} \tag{1}$$

$\mathcal{L}_{REL}$: Relational geometry. For each triple (h, r, t), a relation map ϕ_r predicts the tail from the head:

$$\mathcal{L}_{REL} = d_{\mathcal{M}}(\phi_r(z_h), z_t) + \gamma \cdot \max(0, \Delta - d_{\mathcal{M}}(\phi_r(z_h), z_{t'}))$$

where $z_{t'}$ is a corrupted tail. Unlike TransE, ϕ_r is a full $d \times d$ linear map in the tangent space, enabling non-linear relation modelling.

$\mathcal{L}_{CL}$: Cross-lingual contrastive alignment.

$$\mathcal{L}_{CL} = -\mathbb{E}\left[\log \frac{\exp(-d_{\mathcal{M}}(z_i^{src}, z_i^{tgt})/\tau)}{\sum_j \exp(-d_{\mathcal{M}}(z_i^{src}, z_j^{tgt})/\tau)}\right]$$

This geodesic InfoNCE operates directly in the manifold rather than in a tangent-space cosine proxy. Training on 60K real cross-lingual pairs from Tatoeba (EN-FR, EN-DE, EN-ES) produces genuine multilingual alignment.

$\mathcal{L}_{ENT}$: Entailment and contradiction. Following Vendrov et al. [2016], entailment pairs enforce coordinate-wise order in the tangent space; contradiction pairs are pushed apart by a geodesic margin.

$\mathcal{L}_{COMP}$: Compositional consistency. $\mathcal{L}_{COMP} = d_{\mathcal{M}}(\phi(z_h, r, z_t),\ \phi_r(z_h))$: the compositional operator's output should agree with the relation map's prediction.

$\mathcal{L}_{CM}$: **Cross-modal contrastive.** Same geodesic InfoNCE as $\mathcal{L}_{CL}$, but between image embeddings and text-label embeddings on CIFAR-100.

$\mathcal{L}_{HIER}$: **Hierarchy depth ranking.**

$$\mathcal{L}_{HIER} = \text{MarginRankingLoss}\big(\text{depth}(z_{fine}),\ \text{depth}(z_{coarse}),\ \text{margin} = 1.0\big) + 0.1 \cdot d_{\mathcal{M}}(z_{fine}, z_{coarse})$$

Fine-grained concepts ("pine tree") must be geodesically deeper than their coarse superclass ("tree") by at least margin 1.0, while the proximity term keeps them nearby in the hierarchy.

4.6 Density Estimation: kNN Manifold Density

Rather than a learned energy-based model or VAE (which can collapse on small vocabularies), we use a non-parametric k-nearest-neighbour density estimator in the geodesic metric:

$$\log \hat{\rho}(z_i) = -\log d_{k\text{-NN}}(z_i)$$

where $d_{k\text{-NN}}(z_i)$ is the geodesic distance to the k-th nearest neighbour of z_i in the vocabulary ($k = 10$). High-density regions are where concepts cluster tightly; low-density regions correspond to poorly-attested or novel territory. The density field is normalised to zero mean and unit variance before use in the scoring function.

This makes the logic of ideas explicit: the density field captures where training data provides strong evidence for semantic structure and where the manifold is sparsely populated.

4.7 Inference: Temperature-Controlled Idea Navigation

At inference, the system retrieves concepts appropriate for a query z_q by scoring each vocabulary concept:

$$\text{score}(z_i) = -\frac{d_{\mathcal{M}}(z_q, z_i)}{\tau} + \lambda_\rho \cdot \log \hat{\rho}(z_i)$$

- At $\tau \to 0$: distance dominates; only the nearest manifold neighbour survives. Retrieval is maximally precise.
- At $\tau \to \infty$: distance is suppressed; density dominates. The system returns high-density (well-attested) concepts regardless of query proximity.
- At intermediate τ: a smooth tradeoff between proximity and density, interpretable as the precision/creativity spectrum.

Comparison to LLM temperature. In a language model, temperature scales token logits with no direct semantic interpretation. In the USM, τ operates on geodesic distance between ideas. Low τ means "retrieve exactly what I mean"; high τ means "explore the neighbourhood of what I mean, weighted by what is well-supported." The operation is defined on the semantic manifold, not on a token vocabulary.

5 Implementation

5.1 Architecture

5.2 Training Data

At validation scale, the ConceptNet triples span 85,768 unique concepts across 6 relation types (IS-A, CAUSES, PART-OF, SIMILAR-TO, ANTONYM, CAPABLE-OF), streamed directly from the ConceptNet 5.7 assertions file. At cloud scale, we increase to 400,000 triples covering 209,872 unique concepts.

Table 1: Architecture components. Validation scale uses d=512 and ViT-B/32; cloud scale uses d=1024 and ViT-L/14 (shown).

Component	Architecture	Parameters
Text backbone	`all-MiniLM-L6-v2` (frozen)	22.7M (frozen)
Text projection	Linear(384, 1024) + LayerNorm + $\exp_{\mu_0}$	394,240
Vision backbone	CLIP ViT-L/14 (frozen)	304M (frozen)
Vision projection	Linear(768, 1024) + LayerNorm + $\exp_{\mu_0}$	787,456
Compositional operator	6 relations $\times$ $(W, U, V \in \mathbb{R}^{1024\times1024}, b \in \mathbb{R}^{1024})$	18.9M
Relation maps	6 relations $\times$ $(W \in \mathbb{R}^{1024\times1024}, b \in \mathbb{R}^{1024})$	6.3M
Total trainable		**~26.4M**

Table 2: Training data sources and volumes (validation-scale run; cloud-scale run uses 400K ConceptNet triples).

Stream	Source	Volume	Loss target
Knowledge graph triples	ConceptNet 5.7 (S3 streaming)	100,000	$\mathcal{L}_{REL}$, $\mathcal{L}_{COMP}$
Cross-lingual pairs	Tatoeba (EN-FR, EN-DE, EN-ES)	60,000	$\mathcal{L}_{CL}$
Entailment/contradiction	SNLI	19,978	$\mathcal{L}_{ENT}$
Images with hierarchy labels	CIFAR-100 (100 fine, 20 coarse classes)	50,000	$\mathcal{L}_{CM}$, $\mathcal{L}_{HIER}$

5.3 Training Protocol

Training proceeds in two phases (30 epochs each at validation scale; 60 epochs each at cloud scale):

- **Phase 1** (text only): Trains the text projection, compositional operator, and relation maps using $\mathcal{L}_{REL} + \mathcal{L}_{CL} + 0.5\,\mathcal{L}_{ENT} + 0.1\,\mathcal{L}_{COMP}$. Optimiser: RiemannianAdam [Bécigneul and Ganea, 2019], cosine-annealed from 5×10^{-4} to 10^{-5}.
- **Phase 2** (multimodal): Trains the vision projection at 3×10^{-4} and fine-tunes the text projection at 10^{-5} (100× lower). This joint training allows the text projection to adjust for hierarchy depth while the vision projection aligns images to the text manifold.

Gradient clipping at $\ell_2 = 1.0$ is applied before each step. The Euclidean baseline uses the same architecture, data, epochs, and hyperparameters, with all hyperbolic operations replaced by their Euclidean equivalents.

6 Experiments

We conduct two experiments: a *validation-scale* run that demonstrates the USM's geometric advantages, and a *cloud-scale* run that reveals fundamental optimisation challenges when scaling hyperbolic embeddings.

6.1 Experiment 1: Validation Scale (85K Concepts)

Setup. 100K ConceptNet triples (85,768 unique concepts), d=512, CLIP ViT-B/32, 30+30 epochs on a T4 GPU.

Loss convergence. All losses converge. $\mathcal{L}_{CL}$ decreases steadily, confirming that geodesic-distance InfoNCE produces genuine cross-lingual alignment. The entailment loss converges to near-zero, indicating clean separation of entailment, neutral, and contradiction pairs.

Table 3: Phase 1 training loss convergence (Poincaré model, validation scale).

Loss	Epoch 1	Epoch 10	Epoch 20	Epoch 30
$\mathcal{L}_{total}$	3.456	2.622	2.461	2.374
$\mathcal{L}_{REL}$	1.027	0.872	0.810	0.792
$\mathcal{L}_{CL}$	1.679	1.446	1.404	1.393
$\mathcal{L}_{ENT}$	0.303	0.080	0.053	0.048

Table 4: KG link prediction on held-out ConceptNet triples (validation scale).

Metric	Poincaré	Euclidean	Δ
MRR	**0.329**	0.263	+25.3%
Hits@1	0.037	**0.068**	−45.6%
Hits@10	**0.914**	0.722	+26.6%

KG link prediction. The Poincaré model wins on MRR (+25%) and Hits@10 (+27%). The advantage on neighbourhood-level metrics indicates that curved geometry correctly organises the *region* around each concept.

Table 5: Hierarchy depth accuracy (validation scale).

	Poincaré	Euclidean
Fine > Coarse (%)	**47.0%**	31.0%

Hierarchy depth. The Poincaré model achieves 47% (vs. 31% Euclidean, +16 pp). While neither model reaches the 100% ceiling, the hyperbolic model's advantage demonstrates that curvature provides a natural axis for encoding abstraction level.

Cross-modal retrieval. The Euclidean baseline wins on image-to-text retrieval (R@1: 0.741 vs. 0.579), an early sign that geodesic-based contrastive learning is harder to optimise than its Euclidean counterpart.

Temperature navigation. At low τ, retrieval is semantically coherent and cross-modal: "dolphin" retrieves aquatic mammals and their images; "tree" retrieves specific tree images. At high τ, density-dominated concepts appear alongside query-relevant results, demonstrating the intended precision/creativity tradeoff.

6.2 Experiment 2: Cloud Scale (210K Concepts)

Setup. 400K ConceptNet triples (209,872 unique concepts), d=1024, CLIP ViT-L/14, 60+60 epochs on an NVIDIA A100 (80 GB). All frozen backbone features (CLIP vision, MiniLM text) were pre-computed once and cached on GPU, eliminating the forward-pass bottleneck and allowing batch sizes of 1024.

Results. The results are unambiguously negative for the Poincaré model. Euclidean wins on every metric. Most striking is the hierarchy collapse: the very property that hyperbolic geometry is mathematically designed to capture (depth ordering) regressed to *worse than random* (20% vs. 50% chance), while the Euclidean baseline achieved near-perfect 99%.

Table 6: Cloud-scale results: Poincaré vs. Euclidean on all metrics. Euclidean wins across the board.

Metric	Poincaré	Euclidean	Δ
KG Link Prediction			
MRR	0.010	**0.027**	−63%
Hits@1	0.005	**0.013**	−65%
Hits@10	0.018	**0.054**	−67%
Cross-Modal Retrieval (CIFAR-100)			
R@1	0.688	**0.849**	−16.1 pp
R@5	0.900	**0.967**	−6.7 pp
R@10	0.936	**0.984**	−4.8 pp
Hierarchy Depth			
Fine > Coarse	20.0%	**99.0%**	−79 pp
Mean depth diff	−0.000	+4.073	n/a

Diagnosis.

1. **Radius collapse.** The mean depth difference between fine and coarse concepts is −0.000: all embeddings collapsed to the same radial shell on the Poincaré ball. The exponential map failed to distribute concepts across different depths.
2. **Degenerate relational learning.** The final $\mathcal{L}_{REL} = 0.693 \approx \ln 2$, the value produced by a random 50/50 margin loss. The relation maps stopped learning meaningful structure.
3. **Cross-modal non-convergence.** The Poincaré Phase 2 $\mathcal{L}_{CM}$ converged to 4.437 vs. the Euclidean baseline's 2.950. Geodesic-based InfoNCE in d=1024 has extreme gradient distortion near the ball boundary, preventing the vision projection from aligning with the text manifold.
4. **Euclidean gradient clipping in curved space.** The ℓ_2 gradient clipping (norm ≤ 1.0) clips *Euclidean* gradient norms, but in the Poincaré ball the actual Riemannian gradient magnitude depends on the conformal factor $\lambda(z) = 2/(1 - \|z\|^2)$, which diverges near the boundary. Points near the boundary receive vanishingly small effective updates.

The failure is optimisational, not geometric. The Poincaré ball *can* represent hierarchies with exponentially less distortion than Euclidean space. This is a mathematical fact [Nickel and Kiela, 2017]. The model simply did not find those representations at this scale.

7 Discussion

7.1 The Scaling Gap

The central empirical finding of this paper is a *scaling gap*: hyperbolic embeddings outperform Euclidean at validation scale but collapse at cloud scale. This is not entirely surprising, since prior work on hyperbolic ML has generally demonstrated results at moderate scale [Nickel and Kiela, 2017, Desai et al., 2023, HypStructure, 2024]. However, the magnitude of the collapse (hierarchy accuracy dropping from 47% to 20%, below random chance) is striking and demands explanation.

We attribute the gap to three factors, all optimisational rather than geometric:

1. **Fixed curvature at high dimension.** At c=1.0 and d=1024, the conformal factor $\lambda(z) = 2/(1 - \|z\|^2)$ creates extreme gradient distortion. Points near the boundary (where specific concepts should live) receive vanishingly small effective updates. MERU [Desai et al., 2023] addresses this with learnable curvature initialised near zero; we did not.
2. **Euclidean gradient clipping in curved space.** Standard ℓ_2 gradient clipping treats all parameters equally, but Riemannian gradients have position-dependent magnitude. Proper Riemannian gradient clipping should operate in the tangent space with the metric tensor.

3. **No training curriculum.** We trained all losses jointly from epoch 1. A curriculum that first establishes radial depth ordering (hierarchy) before introducing cross-modal contrastive alignment would likely prevent the observed radius collapse.

7.2 What the USM Extracts That Transformers Do Not

Despite the scaling failure, the *framework* remains well-motivated. The validation-scale results confirm that hyperbolic geometry can encode:

- **Explicit hierarchy** as measurable radial depth: $d_{\mathcal{M}}(z_{\text{retriever}}, \mathbf{0}) > d_{\mathcal{M}}(z_{\text{dog}}, \mathbf{0}) > d_{\mathcal{M}}(z_{\text{animal}}, \mathbf{0})$.
- **Explicit typed relations** as tangent-space maps ϕ_r with per-relation algebraic structure.
- **Explicit density** via kNN geodesic distance, providing per-concept epistemic confidence.
- **Explicit composition** via a bilinear operator with structurally enforced symmetry properties.

These properties are measurable, enforceable via training objectives, and available at inference time as first-class geometric quantities. Transformers encode the same knowledge only as implicit statistical shadows in attention patterns.

7.3 Open Problems and Future Work

Scaling hyperbolic optimisation. The most urgent open problem. We propose three concrete directions: (1) learnable curvature initialised at $c \approx 0.1$ and annealed upward; (2) Riemannian-aware gradient clipping in tangent space; (3) curriculum training that establishes depth ordering before cross-modal alignment.

Integration with LLMs. The USM represents ideas but does not yet produce text. A natural next step is using USM as an external semantic memory for retrieval-augmented generation, or training lightweight adapters from LLM hidden states into USM space. If structured hyperbolic representations improve downstream reasoning, retrieval, or hallucination detection, the practical case for idea-first geometry becomes much stronger.

Deeper hierarchies. CIFAR-100 has only two levels of hierarchy. Evaluation on WordNet (10+ levels) or Wikidata (millions of entities with deep taxonomy) would provide a more demanding test of whether curvature advantages scale with hierarchy depth.

Path to idea-first generation. The complete pipeline has three steps: (1) perception (encode inputs onto the manifold), (2) reasoning (navigate via relation maps and compositional operators), (3) expression (decode manifold positions to surface forms). The scaling challenge reported here sits squarely in step 1, getting concepts onto the manifold correctly. Steps 2 and 3 remain architecturally defined but empirically untested.

8 Conclusion

We have presented the Universal Semantic Manifold, a Riemannian framework for idea-first generation that represents the statistical logic of ideas as explicit geometry on the Poincaré ball.

Our results tell two stories. At validation scale, hyperbolic geometry captures semantic structure that flat geometry cannot: +25% MRR on knowledge graph link prediction, +27% Hits@10, and +16 pp on hierarchy depth accuracy demonstrate that the shape of the representation space matters for the logic of ideas. At cloud scale, naive application of the same training recipe fails entirely: the Euclidean baseline wins on every metric, and the Poincaré model's hierarchy ordering collapses below random chance.

We report both results honestly. The validation-scale wins confirm that the geometric thesis is sound. The cloud-scale failure reveals that realising that thesis at scale requires optimisation techniques, including learnable curvature, Riemannian gradient control, and curriculum training, that are not yet part of the standard hyperbolic ML toolkit. Bridging this gap is the central open problem.

The broader claim remains: current generative AI learns the statistics of tokens. The USM aims to learn the statistics of ideas, including which concepts relate to which, how they compose, how abstract or specific they are, and how well-attested they are. The former produces fluent text. Whether the latter can produce grounded thought depends on solving the scaling problem we have identified.

All code, notebooks, and this paper are publicly available at `https://github.com/AdamVanss/The-USM`.

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